

Flow Matching · arXiv 2210.02747

Scientific Reviewer

8+3 · seat 3

Strengths + actionable cite hygiene

Sep 23 paper club · VLA Architectures

Spine: OT-CFM objective + path design for action statistics

Lipman et al. · Meta AI / Weizmann · pick a path, regress the VF

Strengths · what holds up

Unbiased CFM grads

Thm 2: same gradients as marginal FM under positivity

Unifies diffusion

VE/VP as special cases of Gaussian conditional paths

OT is principled

Displacement interpolant; straight / constant-direction

Multi-metric gains

NLL + FID + NFE together
on matched U-Nets

Off-the-shelf ODE

dopri5 / midpoint enough;
no custom diffusion sampler

Conditional Flow Matching · the objective card

Thm 2: $\nabla \text{CFM} = \nabla \text{FM}$ · never need the marginal VF

Intractable FM

$$L_{FM} = E_{t, p_t} \|v_t(x) - u_t(x)\|^2$$

needs marginal generating field u_t

hard — mix over all data

Tractable CFM ★

$$L_{CFM} = E_{t, q, x} \|v_t - u_t(x|x_1)\|^2$$

per-sample conditional field

same gradients (Thm 2)

OT-CFM recipe (OpenPI / π_0 action expert)

1. Sample path

$$\begin{aligned} t &\sim U(0,1) \\ \varepsilon &\sim N(0, I), \quad a \sim \text{data} \\ a_t &= (1-t)\varepsilon + t a \end{aligned}$$

2. Target VF

$$\begin{aligned} u &\approx a - \varepsilon \\ &\text{(constant direction)} \\ \sigma_{\min} &\text{softens endpoint} \end{aligned}$$

3. Regress

$$\begin{aligned} &\|v_{\theta}(a_t, t | o) - u\|^2 \\ &\text{condition on obs/VLM} \\ &\rightarrow \text{action-chunk expert} \end{aligned}$$

Cite hygiene · actionable caveats

EDM-class recipes

Verify vs modern diffusion recipes before claiming SOTA-style wins

CIFAR FID caveat

Table 1 CIFAR weaker than contemporaneous diffusion lit; arch mismatch

IN-128 gate

Gate SOTA claim — large batch / compute asymmetry vs older GANs

Robotics not in paper

π_0 / BEHAVIOR are post-hoc evidence — not results of Lipman'22

FID $\neq$ accuracy

SR: better FID/IS, lower PSNR/SSIM — pick the right metric

Rectified Flow

Cite Liu et al. concurrency / interpolants — framing+scale, not unique loss

Reviewer checklist

1. Toy OT-CFM first — don't jump to ImageNet claims
2. Confirm π_0 path = OT (not VP) before citing "FM > SM" for robots
3. Separate "FM>SM same U-Net" from "FM solves robot multimodality"
4. Report NFE + FID sample count honesty (10k $\neq$ 50k)
5. Cite Rectified Flow / interpolants as concurrent
6. Never equate BEHAVIOR Σ / N=15 with Lipman defaults

Model	CIFAR-10			ImageNet 32×32			ImageNet 64×64			Model	ImageNet 128×128	
	NLL ₁	FID ₁	NFE ₁	NLL ₁	FID ₁	NFE ₁	NLL ₁	FID ₁	NFE ₁		NLL ₁	FID ₁
<i>Alternatives</i>												
DDPM	3.12	7.48	274	3.54	6.99	262	3.32	17.36	264	MGAN (Huang et al., 2019)	—	58.9
Score Matching	3.16	19.94	242	3.56	5.68	178	3.40	19.74	441	PicGAN2 (Karras et al., 2019)	—	57.5
ScoreFlow	3.09	20.78	428	3.55	14.14	195	3.36	24.95	601	Logo-GAN-AE (Hays et al., 2019)	—	50.9
<i>Ours</i>										Self-cond. GAN (Hays et al., 2019)	—	41.7
FM w/ Diffusion	3.10	8.06	183	3.54	6.37	193	3.33	16.88	187	Uncond. BigGAN (Karras et al., 2019)	—	25.3
FM w/ OT	2.99	6.35	142	3.53	5.02	122	3.31	14.45	138	PGMGAN (Karras et al., 2019)	—	21.7
										FM w/ OT	2.90	20.9

Table 1: Likelihood (BPD), quality of generated samples (FID), and evaluation time (NFE) for the same model trained with different methods.

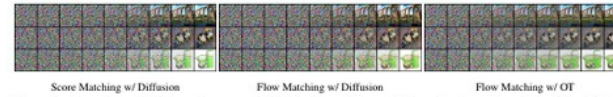


Figure 6: Sample paths from the same initial noise with models trained on ImageNet 64×64. The OT path reduces noise roughly linearly, while diffusion paths visibly remove noise only towards the end of the path. Note also the differences between the generated images.

6.1 DENSITY MODELING AND SAMPLE QUALITY ON IMAGENET

We start by comparing the same model architecture, *i.e.*, the U-Net architecture from Dhariwal & Nichol (2021) with minimal changes, trained on CIFAR-10, and ImageNet 32/64 with different popular diffusion-based losses: DDPM from (Ho et al., 2020), Score Matching (SM) (Song et al., 2020b), and Score Flow (SF) (Song et al., 2021); see Appendix E.1 for exact details. Table 1 (left) summarizes our results alongside these baselines reporting negative log-likelihood (NLL) in units of bits per dimension (BPD), sample quality as measured by the Fréchet Inception Distance (FID; Heusel et al. (2017)), and averaged number of function evaluations (NFE) required for the adaptive solver to reach its a prespecified numerical tolerance, averaged over 50k samples. All models are trained using the same architecture, hyperparameter values and number of training iterations, where baselines are allowed more iterations for better convergence. Note that these are *unconditional* models. On both CIFAR-10 and ImageNet, FM-OT consistently obtains best results across all our quantitative measures compared to competing methods. We are noticing a higher that usual FID performance in CIFAR-10 compared to previous works (Ho et al., 2020; Song et al., 2020b; 2021) that can possibly be explained by the fact that our used architecture was not optimized for CIFAR-10.

Secondly, Table 1 (right) compares a model trained using Flow Matching with the OT path on ImageNet at resolution 128×128. Our FID is state-of-the-art with the exception of IC-GAN (Casanova et al., 2021) which uses conditioning with a self-supervised ResNet50 model, and therefore is left out of this table. Figures 11, 12, 13 in the Appendix show non-curated samples from these models.

Faster training. While existing works train diffusion models with a very high number of iterations (*e.g.*, 1.3m and 10m iterations are reported by Score Flow and VDM, respectively), we find that Flow Matching generally converges much faster. Figure 5 shows FID curves during training of Flow Matching and all baselines for ImageNet 64×64; FM-OT is able to lower the FID faster and to a greater extent than the alternatives. For ImageNet-128 Dhariwal & Nichol (2021) train for 4.36m iterations with batch size 256, while FM (with 25% larger model) used 500k iterations with batch size 1.5k, *i.e.*, 33% less image throughput; see Table 3 for exact details. Furthermore, the cost of sampling from a model can drastically change during training for score matching, whereas the sampling cost stays constant when training with Flow Matching (Figure 10 in Appendix).

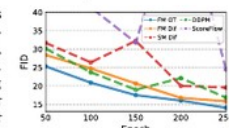


Figure 5: Image quality during training, ImageNet 64×64.

6.2 SAMPLING EFFICIENCY

For sampling, we first draw a random noise sample $x_0 \sim \mathcal{N}(0, I)$ then compute $\phi_1(x_0)$ by solving equation 1 with the trained VF, v_t , on the interval $t \in [0, 1]$ using an ODE solver. While diffusion

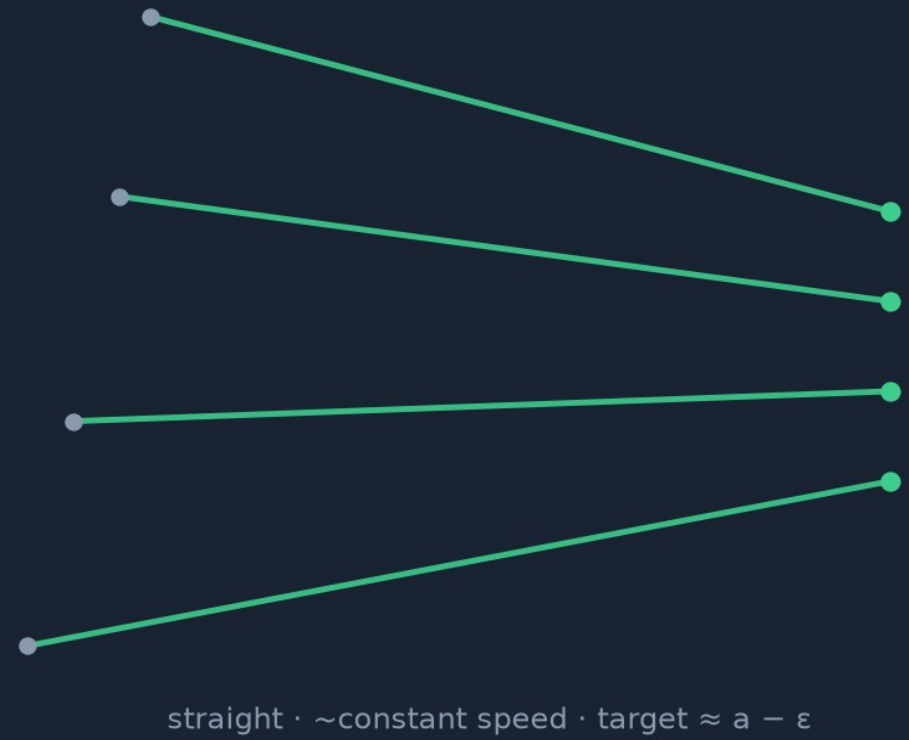
OT straight paths vs diffusion curved paths

Paper Figs 2-3 · constant-direction VF · fewer NFEs for same error

Diffusion / VP path



OT displacement path



Don't overclaim

DP $\neq$ FM

Diffusion Policy = action-space DDPM. Conceptual bridge only. Likely no Lipman cite.

Σ / N=15 $\neq$ Lipman

Correlated noise + multi-sample CFM are BEHAVIOR specializations, not paper defaults.

IN FID $\neq$ robot KPI

ImageNet tables don't transfer to success rate / Hz / contact richness.

Conditional OT $\neq$ marginal OT

Authors admit marginal VF need not be W_2 -optimal — branding can overclaim.

Concurrent literature

Rectified Flow / interpolants overlap — cite them; claim framing+scale+OT emphasis.

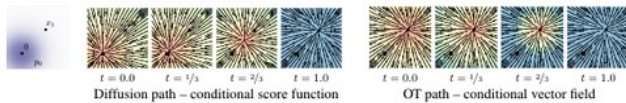


Figure 2: Compared to the diffusion path’s conditional score function, the OT path’s conditional vector field has constant direction in time and is arguably simpler to fit with a parametric model. Note the blue color denotes larger magnitude while red color denotes smaller magnitude.

which, in contrast to the diffusion conditional VF (equation 19), is defined for all $t \in [0, 1]$. The conditional flow that corresponds to $u_t(x|x_1)$ is

$$\psi_t(x) = (1 - (1 - \sigma_{\min})t)x + tx_1, \quad (22)$$

and in this case, the CFM loss (see equations 9, 14) takes the form:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, q(x_1), p(x_0)} \left\| v_t(\psi_t(x_0)) - \left(x_1 - (1 - \sigma_{\min})x_0 \right) \right\|^2. \quad (23)$$

Allowing the mean and std to change linearly not only leads to simple and intuitive paths, but it is actually also optimal in the following sense. The conditional flow $\psi_t(x)$ is in fact the Optimal Transport (OT) *displacement map* between the two Gaussians $p_0(x|x_1)$ and $p_1(x|x_1)$. The OT *interpolant*, which is a probability path, is defined to be (see Definition 1.1 in McCann (1997)):

$$p_t = [(1 - t)\text{id} + t\psi]_* p_0 \quad (24)$$

where $\psi : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the OT map pushing p_0 to p_1 , id denotes the identity map, i.e., $\text{id}(x) = x$, and $(1 - t)\text{id} + t\psi$ is called the OT displacement map. Example 1.7 in McCann (1997) shows, that in our case of two Gaussians where the first is a standard one, the OT displacement map takes the form of equation 22.

Intuitively, particles under the OT displacement map always move in straight line trajectories and with constant speed. Figure 3 depicts sampling paths for the diffusion and OT conditional VFs. Interestingly, we find that sampling trajectory from diffusion paths can “overshoot” the final sample, resulting in unnecessary backtracking, whilst the OT paths are guaranteed to stay straight.

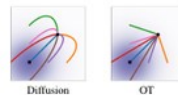


Figure 3: Diffusion and OT trajectories.

Figure 2 compares the diffusion conditional score function (the regression target in a typical diffusion methods), i.e., $\nabla \log p_t(x|x_1)$ with p_t defined as in equation 18, with the OT conditional VF (equation 21). The start (p_0) and end (p_1) Gaussians are identical in both examples. An interesting observation is that the OT VF has a constant direction in time, which arguably leads to a simpler regression task. This property can also be verified directly from equation 21 as the VF can be written in the form $u_t(x|x_1) = g(t)h(x|x_1)$. Figure 8 in the Appendix shows a visualization of the Diffusion VF. Lastly, we note that although the conditional flow is optimal, this by no means imply that the marginal VF is an optimal transport solution. Nevertheless, we expect the marginal vector field to remain relatively simple.

5 RELATED WORK

Continuous Normalizing Flows were introduced in (Chen et al., 2018) as a continuous-time version of Normalizing Flows (see e.g., Kobayev et al. (2020); Papamakarios et al. (2021) for an overview). Originally, CNFs are trained with the maximum likelihood objective, but this involves expensive ODE simulations for the forward and backward propagation, resulting in high time complexity due to the sequential nature of ODE simulations. Although some works demonstrated the capability of CNF generative models for image synthesis (Grathwohl et al., 2018), scaling up to very high dimensional images is inherently difficult. A number of works attempted to regularize the ODE to be easier to solve, e.g., using augmentation (Dupont et al., 2019), adding regularization terms (Yang & Karniadakis, 2019; Finlay et al., 2020; Onken et al., 2021; Tong et al., 2020; Kelly et al., 2020), or stochastically sampling the integration interval (Du et al., 2022). These works merely aim to regularize the ODE but do not change the fundamental training algorithm.

Scientific Reviewer · close / Q&A

Strengths hold; gate the brags; cite concurrency.

Spine reminder:

FM = pick a path, regress the VF; OT → straighter/faster
→ DP chunks → π_0 OT-CFM head → BEHAVIOR fixes robot stats

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